

Utshab Kumar Ghosh

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Education

Ph.D. in Computer Science

Missouri University of Science & Technology

Supervisor: Dr. Shubham Chatterjee

Research Domain: Large Language Models, Information Retrieval, Retrieval-Augmented Generation, Real-World AI Applications.

Result (CGPA/4.0): 4.0

Missouri, USA

Aug 2025 - Present

B.Sc. in Computer Science & Engineering

Rajshahi University of Engineering & Technology

Result (CGPA/4.0): 3.22 (3.48 in last 3 semesters)

Thesis Title: Most Dominant Metabolomic Biomarkers Identification for Lung Cancer

Rajshahi, Bangladesh

Jan 2017 - Oct 2022

Publications

Conference Papers

- Ghosh, U. K. et al. (2026). Reproduction Beyond Benchmarks: ConstBERT and ColBERT-v2 Across Backends and Query Distributions. *ACM SIGIR (A* – Top IR Venue)*.
- Mukhopadhyay, D.[†], Ghosh, U. K.[†], & Chatterjee, S. (2025). Single Turn LLM Reformulation Powered Multi-Stage Hybrid Re-Ranking for Tip-of-the-Tongue Known-Item Retrieval. *TREC ToT Track*.

[†] denotes equal contribution

Journal Articles

- Rifaat, N.[†], Ghosh, U. K.[†], & Sayeed, A. (2022). Accurate gait recognition with inertial sensors using a new FCN-BiLSTM architecture. *Computers and Electrical Engineering (Q1)*, 104, 10842.
- Ghosh, U. K. et al. (2022). Most dominant metabolomic biomarkers identification for lung cancer. *Informatics in Medicine Unlocked (Q2)*, 28, 100824.

Industry Experience

Machine Learning Engineer

Nascenia Ltd.

Dhaka, Bangladesh

Dec 2022 - Jun 2025

- **RAG System Architecture:** Architected a Retrieval-Augmented Generation (RAG) pipeline using LangChain, LlamaIndex, and ChromaDB. Performed R&D on other LLMs (such as Mistral and Bloom) before deploying Llama-3 for document-driven Q/A.
- **Infrastructure Optimization:** Fine-tuned and quantized Llama-3.1 8B models (using 5-bit Quantization), reducing cloud inference costs by almost 70% while maintaining high accuracy.
- **Scale & Impact:** Deployed the solution on AWS with GPU acceleration, handling 10,000+ daily requests and improving citizen service response times by at least 50%.
- **Project Leadership & Client Engagement:** Built and managed high-value client relationships with the Government of Bangladesh and international clients in Turkey and Sweden. Successfully led multiple projects involving teams of 4–10 developers.
- **Full-Stack Integration:** Integrated the AI core into a secure Laravel web application, implementing interactive dashboards and admin panels for government use.
- **IoT Engineering:** Developed and released an Android app for industrial hardware sensors using the Modbus protocol, enabling real-time data collection and monitoring.

Co-Founder & Mentor

Code Studio / Code Studio Academy

Dhaka, Bangladesh

May 2019 – Jun 2025

- **Scaling & Growth:** Founded and grew a tech startup from 2 founders to 15 employees serving 60+ corporate clients. Managed \$40,000 annual operating budget and achieved sustainable revenue through diversified client services.
- **Educational Business Model:** Developed and launched a Machine Learning Bootcamp curriculum; trained 120+ students to address local talent shortages.
- **Impact:** Facilitated career placement for 30+ graduates at top companies in Bangladesh, with about 25% placement rate. Our startup has reached the milestone of over 100 projects so far.
- **Brand Building:** Created technical content reaching 17,000 viewers on YouTube, which has been establishing brand authority.

Medical Data R&D: Developed a computer vision system for Parkinson’s disease detection (0.98 accuracy) using gait analysis of the patients.

Key Projects

Biomedical RAG Prototype for Stem Cells Ongoing

Focus: Retrieval-Augmented Generation. **Contribution:** Designing a prototype system that integrates retrieval from trusted biomedical corpora with LLM generation to deliver accurate, citation-backed answers for stem cell research queries.

CultureQA-LR: Low-Resource Benchmark Ongoing

Problem: LLMs often fail to grasp cultural nuances in low-resource languages. **Solution:** Creating the first culturally specific QA benchmark (focusing on Bangla) with multi-format questions to evaluate LLMs on local history, practices, and reasoning.

CS-PROB: CS Evaluation Benchmark 2025

Problem: Lack of rigorous evaluation for university-level CS tasks. **Solution:** Developing a comprehensive problem suite with detailed rubrics to benchmark LLMs on algorithmic reasoning, code execution, and multi-step problem solving.

NER & POS Tagging for Bangla 2024

Problem: Scarcity of NLP tools for low-resource languages. **Solution:** Built a lightweight FastAPI microservice using optimized deep learning models for low-latency inference.

Bone Marrow Cell Classification 2024

Problem: Manual identification of abnormal blood cells is error-prone. **Solution:** Fine-tuned EfficientNetB7 for high-precision (0.84-0.90) detection of specific artifacts to assist diagnostic workflows.

End-to-End ML Bootcamp Curriculum 2023-2024

Designed and delivered a comprehensive curriculum to bridge the gap between academic theory and industry needs, mentoring 120+ students through the full ML lifecycle.

Machine Learning Competitions

- **Top 10:** *TREC Tip-of-the-Tongue: The phenomenon of failing to recall something from memory, combined with partial recall and the feeling that recall is imminent.*
- **Top 2.6%** (54th/2034): *HackerEarth ML Challenge: A Fine Windy Day.*
- **Top 3%** (86th/2796): *HackerEarth ML Challenge: Fast, Furious and Insured.*
- **Top 12.5%** (443th/3545): *Kaggle PetFinder.my - Pawpularity Contest.*
- **Top 48%** (1019th/2109): *Kaggle TensorFlow - Help Protect the Great Barrier Reef.*
- **Running:** *DataHack Traffic Forecasting.*

Peer Review & Talks

Peer Reviewer 2025-Present

LREC 2026

Speaker: Opportunities in programming career May 2023

IEEE RUET RAS Student Branch Chapter

Skills

Programming	Python, C, OOP, Data Structures & Algorithms.
Data Science, ML and DL	Data Analysis & Visualization, ANN, CNN, RAG, LLM, Information Retrieval, Query Reformulation, Re-ranking, Supervised & Unsupervised Learning, Pipelines, Feature Engineering, Selection & Extraction, Fine-tuning, Training, Prompt Engineering, Quantization, GPU Acceleration.
Frameworks and Tools	TensorFlow, PyTorch, Keras, OpenCV, NumPy, Pandas, Scikit-Learn, LangChain, LlamaIndex, OpenAI API, Hugging Face.
Mathematics for ML	Algebra, Statistics, Probability, Geometry, Matrix.
Web Development	ReactJS, SaaS, Tailwind, JavaScript, Django, Flask.
Database	MySQL, MariaDB, ChromaDB, Redis, Firebase, MongoDB.
Cloud	AWS, Azure, Apache Server, Nginx.
Business & Leadership	Product Development, Client Relationship, Team Management, Mentorship.
Miscellaneous	Git, Docker, Linux, Gradio, Latex, MS Office, Adobe Illustrator, Premiere Pro.

Accomplishments

- **The Swedish Institute Scholarship for Global Professionals, 2025** (awarded to 300 scholars worldwide).
- **Kaggle Expert (x3)**: Demonstrated consistent performance across multiple ML competitions.
- **300+ Problems solved** in *Codeforces*, *Codechef*, *Leetcode*.
- **Gold Medalist** in *National High School Programming Contest*.
- **Deep Learning Specialization** in *deeplearning.ai*.
- **Applied Data Science Specialization** in *IBM*.
- **Data Scientist Career Track** in *Datacamp*.
- **AI for Medical Diagnosis** in *deeplearning.ai*.

Activities

Cultural Secretary

Bangladesh Students' Association, Missouri S&T

2025 – Present

- Orchestrating university-wide cultural events.
- Managing event budgets, logistics.
- Coordinating volunteer teams for community engagement.

Vice President (Admin)

Onuronon, RUET

2021 – 2022

- Led administrative operations for the university's only cultural club.
- Increased member engagement by 40% through targeted recruitment and organizing multidisciplinary events.

Tech Writer

Medium

- Writing technical articles on SWE and ML to simplify complex concepts for developers.

References

Available upon request.